Parametric Equations

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Eliminating the parameter

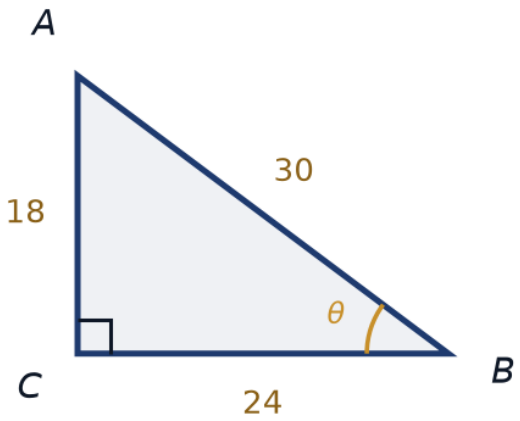
Consider the curve $x = 2t + 1$, $y = t^2$.

a Eliminate the parameter to write y as a function of x.

b Identify the curve.

2 [Foundational] • Target: Eliminating the parameter

Eliminate the parameter for $x = 3\cos t$, $y = 3\sin t$, $0 \le t < 2\pi$, and describe the curve.



3 [Foundational] • Target: Eliminating the parameter

For the line $x = t - 2$, $y = 2t + 3$:

a Find the points corresponding to $t = 0$ and $t = 2$.

b Eliminate the parameter to write the Cartesian equation.

4 [Foundational] • Target: Projectile motion

A projectile follows $x = 40t$, $y = -5t^2 + 30t$ (metres, seconds).

a When does the projectile land ($y = 0$, $t > 0$)?

b Find the horizontal range.

c Find the maximum height.

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5 [Foundational] • Target: Projectile motion

A second projectile is launched with $x = 25t$, $y = -5t^2 + 40t$. Find its time of flight and maximum height.

6 [Foundational] • Target: Ellipses and other parametric curves

Eliminate the parameter for $x = 4\cos t$, $y = 2\sin t$, and identify the resulting curve.

7 [Foundational] • Target: Ellipses and other parametric curves

Eliminate the parameter for $x = t^2 - 1$, $y = t^3$ by expressing t in terms of x where possible, or by finding a relation between x and y directly.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Finding the parametric derivative

For the curve $x = t^2$, $y = 4t$, find $\frac{dy}{dx}$ in terms of t using $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$.

9 [Intermediate] • Target: Finding the parametric derivative

For the curve $x = \cos t$, $y = \sin t$, find the slope of the tangent line at $t = \frac{\pi}{4}$.

10 [Intermediate] • Target: More parametric practice

A Ferris wheel of radius 10 m has its center 12 m above the ground and completes one revolution every 40 seconds. Write parametric equations for the height y of a rider above the ground as a function of time t (seconds), assuming the rider starts at the bottom of the wheel.

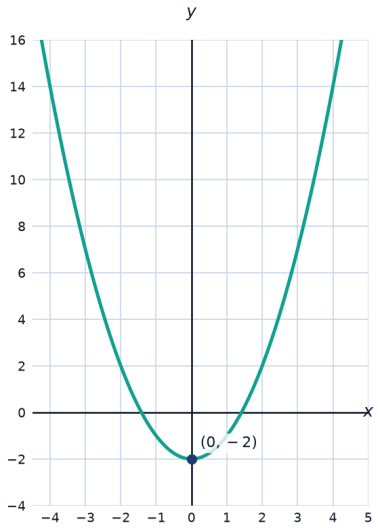
11 [Intermediate] • Target: More parametric practice

Two particles travel along paths $x_1 = t,\ y_1 = t^2$ and $x_2 = 4 - t,\ y_2 = 3t - 2$, sharing the same parameter t (time). Find the value of t at which they collide, and the point of collision.

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12 [Intermediate] • Target: Graphing a parametric curve after elimination

For $x = t$, $y = t^2 - 2$ (with t ranging over all reals), the eliminated Cartesian form $y = x^2 - 2$ is shown. Use the graph to describe the direction the curve is traced as t increases.



13 [Intermediate] • Target: Restricting the parameter domain

The curve $x = \cos t$, $y = \sin t$ for $t \in [0, \pi]$ traces only part of the unit circle. Describe which part, and state the domain and range restrictions this implies on x and y.

14 [Intermediate] • Target: Parametrizing a line segment

Write parametric equations for the line segment from $A(2, -1)$ to $B(6, 5)$, using $t \in [0, 1]$ (so $t = 0$ gives A and $t = 1$ gives B).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Parametrizing a line segment • Cross-Lesson: 1.03 Average Rate of Change

Using the parametrization above, find the point at $t = 0.25$ (one-quarter of the way from A to B).

16 [Challenging] • Target: Parametric speed and motion review • Cross-Lesson: 2.01 Right-Triangle Trigonometry

A particle's position is $x(t) = 3t$, $y(t) = 4t$ for $t \geq 0$ (measured in meters, seconds). Find the particle's speed, and confirm it matches $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$.

17 [Challenging] • Target: Parametrizing a line segment • Cross-Lesson: 2.02 Radian Measure and Arc Length

Eliminate the parameter t from the parametric curve $x(t) = 3 + 4\cos t$, $y(t) = -2 + 5\sin t$ for $0 \leq t \leq 2\pi$. Identify the conic section, its center, and its vertices.

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18 [Challenging] • Target: Parametric speed and motion review • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Find the standard equation $Ax + By = C$ of the line perpendicular to $3x - 4y = 12$ that passes through the midpoint of the line segment joining $P(-2, 7)$ and $Q(6, 1)$.

19 [Challenging] • Target: Eliminating the parameter • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Find the exact area of the region enclosed between the curve $y = 4x - x^2$ and the line $y = x$.

20 [Challenging] • Target: Projectile motion • Cross-Lesson: 3.01 The Unit Circle and Exact Values

A curve has equation $y = 2x^3 - 9x^2 + 12x + 1$. (a) Find the coordinates of both stationary points. (b) Determine the nature of each stationary point using the second derivative $\frac{d^2y}{dx^2}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Eliminating the parameter

A particle of mass 6 kg moving at 6 m/s collides and sticks to a particle of mass 6 kg moving at 0 m/s (same line of motion). Find the common velocity after collision.

a. 2

b. 4

c. 6

d. 3

22 [Foundational] • Target: Projectile motion

A particle moves in a straight line such that its displacement s metres from a fixed origin after t seconds is given by the equation $s = t^3 - 6t^2 + 9t + 2$. The velocity of the particle is given by $v = \frac{ds}{dt}$. Find the value of t for which the particle is instantaneously at rest and then determine the displacement of the particle at that instant.

a. t=1 s, s=6 m

b. t=2 s, s=4 m

c. t=3 s, s=2 m

d. t=1 s, s=2 m

23 [Foundational] • Target: Ellipses and other parametric curves

The region R in the first quadrant is bounded by the curve $y = \frac{6}{\sqrt{3x+4}}$, the coordinate axes, and the vertical line $x = k$, where $k > 0$. Given that the total area of R is 8, and the vertical line $x = c$ (where $0 < c < k$) divides R into two regions of equal area, find the exact value of c.

a. $\frac{5}{3}$

b. 2

c. $\frac{7}{3}$

d. $\frac{13}{27}$

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24 [Foundational] • Target: Finding the parametric derivative

The curve C has equation $y = \frac{ax + 6}{x^2 + 3}$ where a is a constant. Given that C has a stationary point at $x = 1$, find the coordinates of the other stationary point of C.

a. $(-3, -1)$

b. $(-3, 1)$

c. $(3, 1)$

d. $(-3, -2)$

25 [Foundational] • Target: More parametric practice

A storage bin contains red, blue, and green blocks. The ratio of red blocks to blue blocks is $3 : 5$. After 12 green blocks are added to the bin, with no other blocks added or removed, the total number of blocks in the bin is 60, and the probability of randomly selecting a green block from the bin is $\frac{1}{3}$. How many red blocks were in the bin originally?

a. 12

b. 15

c. 18

d. 25

26 [Foundational] • Target: Graphing a parametric curve after elimination

In the diagram below, line segment DE is parallel to BC. Given that AD = 6, DB = 4, and DE = 12, find the length of BC.

a. 18

b. 20

c. 30

d. 15

27 [Foundational] • Target: Restricting the parameter domain

A particle moves along a straight line. Its velocity, v m/s, at time t seconds is given by $v = 3t - 2$. Find the displacement of the particle during the first 4 seconds of its motion.

a. 16 m

b. 20 m

c. 24 m

d. 28 m

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28 [Foundational] • Target: Parametrizing a line segment

A particle has initial velocity $u = 3$ m/s and constant acceleration $a = 3$ m/s^2. Find its speed after travelling a displacement $s = 11$ m.

a. $\sqrt{79}$

b. $\sqrt{71}$

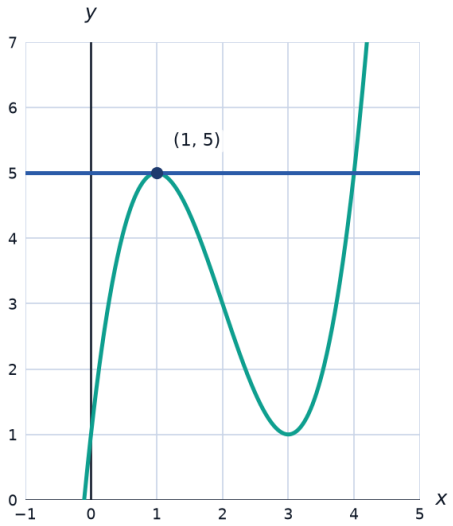
c. 75

d. $\sqrt{75}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Parametric speed and motion review

Consider the function $f(x) = x^3 - 6x^2 + 9x + 1$. A line tangent to the graph of $f(x)$ at $x = 1$ intersects the graph at another point P. Determine the x-coordinate of point P.



a. 4

b. 5

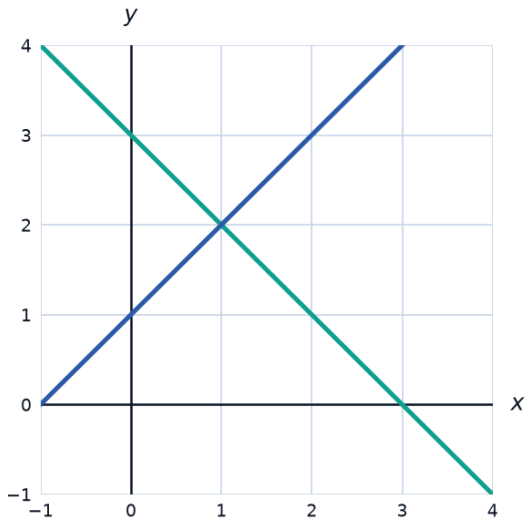
c. 3

d. 2

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30 **[Intermediate]** • **Target: Eliminating the parameter**

The diagram shows the graphs of two linear equations. The point of intersection of the two lines represents the solution to the system of simultaneous equations. Find the coordinates of the point of intersection.



a. (1, 2)

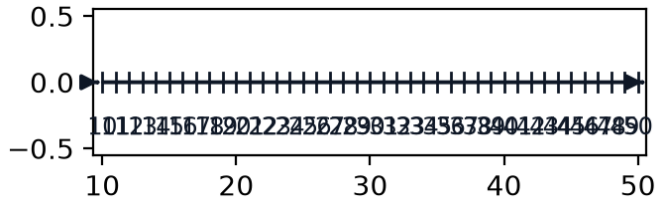
b. (2, 1)

c. (3, 0)

d. (0, 3)

31 **[Intermediate]** • **Target: Projectile motion**

The closed interval $[12, 48]$ is shown on the number line below. How many integers x in this interval satisfy the linear congruence $2x \equiv 1 \pmod 5$?



a. 6

b. 7

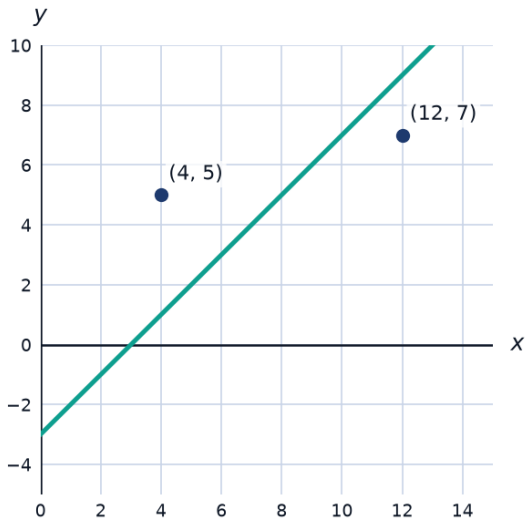
c. 8

d. 9

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32 **[Intermediate]** • **Target: Ellipses and other parametric curves**

The graph of the exponential function $f(x) = a \cdot \log_3(x - h) + k$ is shown below. The function has a vertical asymptote at $x = 3$ and passes through the points $(4, 5)$ and $(12, 7)$. Determine the value of the parameter a.



a. 1

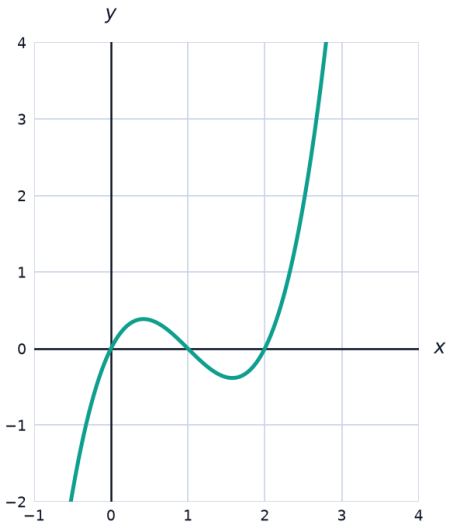
b. 2

c. 1/2

d. 4

33 **[Intermediate]** • **Target: Finding the parametric derivative**

A curve is given by the equation $y = x^3 - 3x^2 + 2x$. Find the gradient of the curve at $x = 2$.



a. 1

b. -2

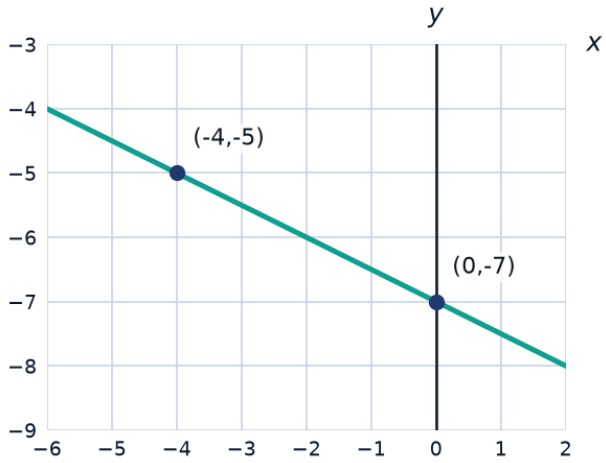
c. 2

d. 0

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34 **[Intermediate]** • **Target: More parametric practice**

Find the equation of the line through $(0, -7)$ and $(-4, -5)$, in the form $y = mx + c$.



a. y=-1/2x+7

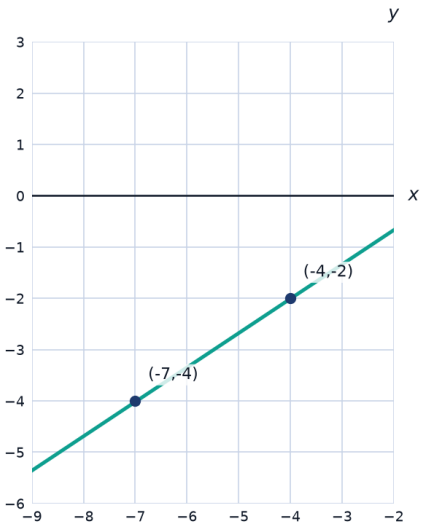
b. y=-1/2x-7

c. y=-1/2x-6

d. y=1/2x-7

35 **[Intermediate]** • **Target: Graphing a parametric curve after elimination**

Find the slope of the line through $(-4, -2)$ and $(-7, -4)$.



a. 2/3

b. -4/3

c. 8/3

d. 13/6

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36 [Intermediate] • Target: Restricting the parameter domain

A particle of mass 2 kg is acted upon by a constant force $\mathbf{F}$. At time $t = 0$, the particle is at the origin $(0, 0)$ and has velocity $\mathbf{v} = (3\mathbf{i} + 4\mathbf{j})$ m/s. At time $t = 2$ s, the particle's velocity is $\mathbf{v} = (7\mathbf{i} + 12\mathbf{j})$ m/s. Determine the magnitude of the constant force $\mathbf{F}$ acting on the particle.

a. $\frac{5\sqrt{17}}{2}$ N

b. $5\sqrt{17}$ N

c. $\frac{5\sqrt{2}}{2}$ N

d. $5\sqrt{2}$ N

37 [Intermediate] • Target: Parametrizing a line segment

In the standard (x, y) coordinate plane, the line $y = mx$ (where $m > 0$) is tangent to the circle $(x - 10)^2 + y^2 = 36$. What is the value of m?

a. $\frac{3}{5}$

b. $\frac{3}{4}$

c. $\frac{4}{5}$

d. $\frac{4}{3}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Parametric speed and motion review • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

A factory produces light bulbs. The probability that a randomly selected light bulb is defective is p. The number of defective light bulbs in a random sample of n bulbs follows a binomial distribution $X \sim B(n, p)$. It is known that the mean number of defective bulbs in a sample of $n = 25$ is 2. \\ (i) Calculate the value of p. \\ (ii) For a sample of $n = 50$ bulbs, calculate the probability of finding exactly k defective bulbs, where k is the smallest integer such that $P(X \le k) > 0.9$. \\ (iii) If a sample of $n = 50$ bulbs is taken, find the smallest value of n' such that the probability of observing at least one defective bulb is greater than 0.999.

a. 0.04

b. 0.08

c. 0.12

d. 0.16

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39 [Challenging] • Target: Eliminating the parameter • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Consider the rational function $f(x) = \frac{x^2 - 2x + c}{x^2 + 2x + c}$ where c is a real constant. If the domain of f is the set of all real numbers and the maximum value of $f(x)$ is 4, what is the value of c?

a. $\frac{25}{9}$

b. $\frac{9}{4}$

c. 3

d. $\frac{16}{5}$

40 [Challenging] • Target: Projectile motion • Cross-Lesson: 3.01 The Unit Circle and Exact Values

In the standard (x, y) coordinate plane, line L is perpendicular to the line $3x + 4y = 15$ and passes through the point $(3, 1)$. Line L intersects the line $2x + y = 17$ at the point (p, q). What is the value of $p + q$?

a. 5

b. 6

c. 8

d. 11

41 [Challenging] • Target: Ellipses and other parametric curves • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

A curve has equation $y = x^2 e^{3x}$. Find the gradient of the curve at the point where $x = 1$.

a. $5e^3$

b. $6e^3$

c. $3e^3$

d. $2e^3$

42 [Challenging] • Target: Finding the parametric derivative • Cross-Lesson: 3.03 Sum and Difference Formulas

A particle of mass 8 kg is moving under the action of a net force $\mathbf{F}$ N. The acceleration of the particle, $\mathbf{a}$ m/s^2, is given by the vector $\mathbf{a} = (2t - 4;\, t^2 - 9)$ for $t \geq 0$, where t is the time in seconds. The particle is initially at rest at the origin.

a. 8 N

b. 16 N

c. 24 N

d. 32 N

43 [Challenging] • Target: More parametric practice • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

The random variable X has a Poisson distribution with parameter λ, where $\lambda > 0$. Given that $\mathrm{P}(X = 3) = \mathrm{P}(X = 4)$, find the value of λ.

a. 4

b. 3

c. 12

d. $\frac{4}{3}$

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44 [Challenging] • Target: Graphing a parametric curve after elimination • Cross-Lesson: 4.01 Circles and Parabolas

The function f is defined by $f(x) = a \cdot b^x + c$, where a, b, and c are constants with $a > 0$, $b > 0$, and $beq1$. In the xy-plane, the horizontal asymptote of the graph of $y = f(x)$ is the line $y = -4$. If the graph passes through the points $(0, -1)$ and $(2, 8)$, what is the value of $f(3)$?

a. 20

b. 24

c. 32

d. 44

45 [Challenging] • Target: Restricting the parameter domain • Cross-Lesson: 4.02 Ellipses

A transformation is applied to a triangle ABC, mapping it to triangle A'B'C'. The transformation can be represented by a matrix multiplication of the form $M(x; y) = (3, 1; 1, k)(x; y)$, where M is the transformation matrix and k is an unknown parameter. If the original point A is $(2, 1)$ and its transformed image A' is $(7, 5)$, find the value of k.

a. 2

b. 3

c. 4

d. 5

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